

STRAT-TRACE

Source-tagged ⁷Be for process-aware aerosol model evaluation

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1. Excellence

1.1 Quality and pertinence of the project’s research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)

STRAT-TRACE will determine when measurements of ⁷Be in air and deposition can distinguish a change in upper-atmospheric supply from a change in wet removal. Controlled DEHM experiments, verified source budgets and tests against observations will establish which explanations can be separated, and where the available measurements remain inconclusive.

A simulation can match measured concentrations for different physical reasons. Using the existing beryllium-7 (⁷Be) implementation in the Danish Eulerian Hemispheric Model (DEHM), this project will provide a reusable test of those explanations. It progresses from verified source budgets (O1), to distinguishing supply and removal changes (O2), to testing the conclusions in independent years and observing conditions (O3).

Production, boundary supply, transport, radioactive decay and deposition jointly determine the measured ⁷Be signal [1–3]. Air concentration and deposition provide complementary evidence. Production uncertainty and rainfall conditions affect what can be inferred [1–4]. Source labels track model contributions; observations test whether their responses can be distinguished despite sampling limits and uncertainty.

State of the art and the specific advance

Established capability	What STRAT-TRACE will add
Joint ⁷ Be concentration–deposition evaluation [1,2] and improved production formulations [3].	A controlled test of when joint observations reject compensating supply–removal explanations, including nuisance-source uncertainty.
Source tagging already exists; GEOS-Chem includes a stratospheric-source Be7s tracer [16].	A verified multi-origin budget linked to discriminatory power at the actual measurement intervals and its dependence on observing regime.
Controlled removal contrasts [5] and transport-model comparisons [6].	A compact within-DEHM experiment set with identical meteorology and independent evaluation, yielding a protocol reusable in other modelling systems.

Table 1. Novelty is the integrated, observation-aware test of process explanations; individual tracer and statistical tools are established.

Synthetic tests will assess whether the observations distinguish halved or doubled wet-removal rate coefficients from compensating supply changes, targeting ≥80% correct recovery at ≤5% false discrimination. Results will identify useful measurement combinations and conditions. DEHM’s contribution to CAMS [17] provides a route for developers to apply the evaluation protocol.

A focused design for a 24 month fellowship

The 24-month project uses the common 75 km DEHM research configuration with reduced chemistry, WRF/ERA5, 29 layers and a top near 100 hPa [7]. AU-confirmed resources support matched experiments over 1986–2024/2025. Three wet-removal settings share meteorology, with variable and fixed production to test source dependence across three complete solar cycles. Runs compared with observations use archived solar and geomagnetic inputs. Sensitivities apply to this configuration; the evaluation protocol can be reused, with validation in finer or operational configurations. Existing implementation and observations make this focused design feasible. Climate-change attribution is outside its scope.

Measurable objectives, hypotheses and success criteria

Objective / delivery	Testable hypothesis and evidence
O1 — Source-resolved response By M9: verified source contributions and budgets.	H1: wet-removal changes produce origin-dependent responses. Compare source fractions and absolute losses across years and solar phases under matched variable/fixed production. Test whether contrasts exceed numerical and sampling uncertainty.
O2 — Discrimination of explanations By M15: supply–removal response library.	H2: joint air–deposition observations discriminate scenarios that air-only evaluation leaves unresolved. Compare recovery at the same controlled false-discrimination rate and quantify the added information from deposition.
O3 — Transfer and observing value By M21: held-out tests and discrimination map.	H3: budget-supported explanations retain joint skill across held-out years and regimes. Fixed-production controls test source dependence. Assess transfer using separate air/deposition scores and pre-specified uncertainty tolerances.

The hypotheses compare wet removal with alternative explanations involving production, upper and lateral boundary supply, and measurement representativeness. Success requires verified response budgets and a reproducible assessment of observational support. Where explanations remain indistinguishable, the benchmark will quantify the tested discrimination limits and the additional observing information needed to separate them.

Existing implementation and observational preparation

Chham and Ye have implemented ^7Be in DEHM. Their EGU 2026 abstract documents production mapping, WRF/ERA5 meteorology, aerosol-bound treatment and surface evaluation [7]. The applicant's preparatory comparison of model resolutions motivates the controlled experiments:

DEHM: 25 km relative to 75 km	Median response
Surface ^7Be concentration	+5.9%
Bulk deposition: wet + dry	+13.6%
Precipitation	−45.9%

Table 2. Applicant's preliminary JJA analysis, 2009–2024, 79 model sampling locations. Each response is $100 \times (\text{fine/coarse} - 1)$ from paired seasonal means, followed by a spatial median. Locations may share grid cells. These model contrasts define tracer supply and removal exposure as processes to test.

Higher deposition alongside lower precipitation points to tracer availability and exposure to removal as explanations to test. These preliminary contrasts do not establish a unique cause or show that either resolution is more accurate. Controlled interventions and independent observations will test the explanations.

The European air archive was obtained primarily through the European Commission's Joint Research Centre (JRC), retaining national-programme attribution [22]. It includes over 54,000 records from 83 named series with checked values, intervals and units in 2009–2024. The joint archive covers 43 European stations with airborne and rainwater ^7Be measurements. Deutscher Wetterdienst (DWD) supplies aerosol, deposition and station/method metadata; 2009–2024 remains the recent joint baseline [20].

Four air series occur in 1986: Berlin, Braunschweig, Risø and Wien-Hohe Warte, with broader coverage from 2006. Chham and Ye have signed a Comprehensive Nuclear-Test-Ban Treaty Organization (CTBTO) agreement to extend air coverage internationally; delivery is pending [23]. WP1 verifies paired windows, station continuity and overlaps; DWD historical recovery continues.

1.2 Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices)

Tracking where ^7Be enters the model

Source labels share transport, mixing, decay and wet/dry deposition. Code, operators, aerosol treatment and CRAC:Be [3,9] are documented and fixed across experiments. Layer 1 is the surface.

Production $P(x, t)$ is divided among non-negative source weights w_k , with $\sum_k w_k = 1$ and $P_k = w_k P$. Labels share transport, decay and deposition. They record where the tracer was produced or entered the model, not its later altitude or last tropopause crossing.

Label	Source partition and interpretation
UPPER / MIDDLE / LOWER	Production within atmospheric cells at $p < 300$ hPa, $300 \leq p < 700$ hPa, and $p \geq 700$ hPa, respectively. These are diagnostic pressure bands.
TOP / LATERAL	Actual upper and lateral exchanges, separate from in-domain production. TOP follows the prescribed upper concentration; boundary additions and withdrawals are budgeted.
INITIAL	Initial inventory propagated through spin-up; its remaining contribution is tracked separately.

Table 3. Six labels and an independent total per production treatment; a short test shifts the pre-specified pressure cut-offs.

The upper concentration $n_{\text{top}} = \frac{Q_{\text{CRAC}}(100 \text{ hPa}, \varphi, t)}{\lambda_7}$ is prescribed in atoms kg^{-1} of air; Q is local production in atoms $\text{kg}^{-1} \text{s}^{-1}$ and λ_7 the decay constant in s^{-1} . This local production–decay balance retains latitude and solar dependence. TOP records actual boundary supply separately from production inside the domain. Production and circulation above the ceiling remain unresolved. UPPER + TOP is not the complete stratospheric contribution: pressure bands and entry routes do not establish stratospheric origin.

Fixed-production runs preserve CRAC:Be’s latitude–altitude structure but hold it constant in time; TOP uses this field divided by λ_7 . LATERAL repeats an annual concentration cycle. Weather-dependent exchanges remain budgeted. Uncontrolled lateral variability limits attribution to in-domain labels and TOP.

Within each wet setting, variable- and fixed-production runs share meteorology, transport and removal operators. Comparisons between matched calendar periods in different years test transport and removal together; comparisons between wet settings test removal. Seasonal lateral inputs remain. Normalised diagnostics are checked against fixed-source runs for spatial and delayed source effects [3].

Checking tracer totals and mass balance

Summed labels must reproduce total concentration, deposition and burden, also after reweighting. Checks cover units, limiters, boundary updates and nested-label transfer; boundary inventory changes stay separate from production. The tolerance is below 0.1% above a declared absolute floor and ten times smaller than the smallest interpreted response.

For each label k and common control volume, the inventory N_k (total atoms) will satisfy:

$$\frac{dN_k}{dt} = P_k + F_{\text{in},k} - F_{\text{out},k} - \lambda N_k - L_{\text{wet},k} - L_{\text{dry},k} + \varepsilon_k.$$

P and F are production and transfer rates, L is removal and ε the residual. At each time step, an internal transfer is recorded as an equal loss and gain, retaining its origin label. Accounting includes changes of pressure band, moving boundaries, boundary inventory adjustments and numerical-operator losses, without counting production twice. The whole-domain budget must close; unresolved local terms are reported.

Verified outputs include concentration C , deposition D , and source fractions $\frac{C_k}{C_{\text{total}}}$ and $\frac{D_k}{D_{\text{total}}}$ with minimum denominators to avoid unstable ratios. Wet losses are integrated by origin and pressure band. The diagnostic $\frac{\int L_{\text{wet}} dt}{\int N dt}$ divides time-integrated wet loss by time-integrated inventory: a removal rate with inverse-time units.

The experiment matrix across three solar cycles

All experiments cover 1986–2024/2025: 39 or 40 years, ending in 2025 where consistent inputs permit, otherwise 2024. They span complete solar cycles 22–24 and the rising and maximum phases of cycle 25 [21]. Meteorology and years are common to all settings. Verified station records from 1986–2005 support temporal tests. The denser later network supports geographic comparisons and joint air–deposition tests, centred on 2009–2024. WP1 checks input continuity and coverage; the contingency in Section 3.1 retains a common period for all experiments.

Case	Intervention	Coverage / interpretation
R + tags	Time-varying CRAC:Be production and linked upper-boundary concentration; reference process operators.	1986–2024/2025. Reference budgets and direct comparison with available observations.
W– / W+ + tags	Multiply the reference aerosol wet-removal rate coefficients by $\alpha_w = 0.5$ or 2; reference $\alpha_w = 1$.	Same full period and production forcing as R. Prescribed stress contrasts test sensitivity; they are not calibrated improvements or confidence limits.
S contrasts	Reweight UPPER and TOP after linearity tests, preserving production–TOP links for source scenarios; TOP-only contrasts test the boundary approximation.	Matched-tag reconstruction over the full period, separately for each source treatment. Supply changes; transport operators do not.
Fixed-source partners	Repeat a common production field and its linked TOP concentration; control LATERAL variability as specified above.	All three wet settings, 1986–2024/2025. Interannual atmospheric response under common source forcing.
Verification controls	Source linearity, concentration/production units, boundary updates, pressure-band and spin-up checks.	Bounded pilots. Spin-up precedes the analysed 1986 start; INITIAL influence is checked.

Table 4. Three wet settings over 39/40 years, each with variable- and fixed-production tracers. Their execution and species costs are accounted for in Section 3.1; source reweighting requires verified linearity.

Wet-removal rate coefficients are scaled before time integration; the amount actually removed enters the deposition budget, with positivity maintained. Within each production treatment, production, prescribed boundary concentrations, source masks, decay, dry deposition, meteorology and transport operators stay identical across wet settings. These contrasts test represented wet-removal strength, not separate in-cloud and below-cloud mechanisms. WP1 verifies where the scaling is applied and archives the operator tests.

Supply tests assess production and lateral-input uncertainty as alternative explanations. Reweighting each wet setting's verified source labels preserves its transport and removal response. Changing CRAC:Be production at 100 hPa also changes TOP through the equilibrium relation; TOP-only changes test that boundary approximation. Comparing the wet-setting contrast in the same output under two supply assumptions—a difference of differences—measures how altered supply changes exposure to removal.

Testing results in independent years

Before fitting, seasons are grouped by rainfall amount, wet-day frequency, persistence and atmospheric mixing. Variable- and fixed-source experiments test wet/dry years and solar phases. Long records support comparisons within stations; the wider network tests geographic transfer. Stable station and measurement-method subsets, common sampling intervals and fixed aggregation weights separate atmospheric changes from changes in network composition. Conclusions retain the geographic and temporal scope of the supporting observations.

Whole-year and multi-year blocks are assigned to fitting and independent evaluation before the analysis, using meteorology, solar phase, coverage and changes in collector method. Every setting covers both sets; runs with realistic production are compared with observations. Joint tests use paired air–deposition intervals. Later records follow a pre-registered extension without retuning. Stations sharing grid cells are grouped; resampling whole years tests dependence on atmospheric conditions. Daily rainfall supports daily/weekly diagnostics, not sub-hourly inference.

Matching measurements and distinguishing explanations

CTBTO is expected to add daily air records from around 80 sites, mainly since 2000, in Europe, North America and Asia. WP1 checks delivered station-years within DEHM coverage and JRC/DWD overlaps using station histories and sampling windows. These data support event-scale concentration tests. DWD records before and after the 2018 collector change are assessed separately [20]. Joint tests use valid pairs from the 43-station European archive; air-only sites test geographic transfer. These counts are separate from the 79 preliminary model locations.

To compare like with like, modelled atoms kg^{-1} are converted to activity using the decay constant and matching air density. Air concentrations are averaged over collection windows in mBq m^{-3} , weighted by sampled airflow where possible. Deposition is integrated over its own windows in Bq m^{-2} , including wet plus dry flux for bulk collectors [20]. Rainwater records in Bq L^{-1} require matching rain depth in L m^{-2} . Method, uncertainty and decay conventions are retained. Collector–grid differences enter representativeness error; nondetections are censored and missing values are not zeros.

Runs with time-varying CRAC:Be production are tested against air measurements, deposition measurements and both together. Fixed-production runs show how conclusions depend on the source. Known scenarios generate synthetic observations with the real sampling intervals, gaps and uncertainties, including cases outside fitted source assumptions. Tests retain measurement and representativeness errors, temporal and spatial dependence, model discrepancy and rainfall-related covariance in converted deposition values.

We first check whether responses differ enough to distinguish after allowing for uncertainty, using scaled sensitivities and their singular values. Likelihood profiles [8] then test source amplitudes and other uncertain terms within each wet setting. The resulting map shows recovery probabilities for prescribed contrasts and detectable source-amplitude bounds. Synthetic tests target $\geq 80\%$ recovery at $\leq 5\%$ false discrimination, with simulation uncertainty. Where these targets are missed, we report measured recovery and the explanations that remain indistinguishable.

Parameters, rainfall/mixing groups and scoring rules are fixed before examining held-out tracer responses. Air and deposition errors, joint predictive scores and interval coverage are compared with R using whole-year blocks and separate collector-method groups. An explanation is supported only if its budget effect is verified and improvement persists in independent years without a compensating deterioration beyond the pre-specified uncertainty tolerance. Otherwise it is rejected or remains indistinguishable among the tested alternatives.

Combining disciplines and sharing results

Environmental radioactivity supplies production, decay and sampling conventions; transport physics supplies source budgets; statistics tests what the measurements can distinguish. Each simulation passes through the sampling calculation and recovery tests, so the physical interpretation is assessed against the information in the observations.

The M3 data-management plan will record data provenance and permissions, budget conventions, metadata, storage needs, retention and maintenance responsibilities during and after the fellowship. JRC/national, DWD and CTBTO records follow their providers' access conditions. WP1 checks transfers and numerical closure by M4. Protocols and evaluation blocks are versioned before fitting. Curated model products, publication-supporting data and reusable code receive DOI archives, suitable licences and FAIR metadata [19]. Restricted observations remain with authorised users; synthetic examples enable independent reuse.

Sex and gender are not explanatory variables for the atmospheric processes studied. The project uses environmental observations and model output, with no human participants, exposure or health-outcome data; this assessment will be revisited if the research changes. Coverage and sampling tests address geographic and observational representativeness. Section 1.3 describes inclusive mentoring. No new radioactive releases or handling are planned.

1.3 Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host

Aarhus University (AU), Department of Environmental Science, Roskilde, will host the project with Zhuyun Ye as main supervisor. Her expertise includes regional Eulerian modelling, data assimilation [13] and the CAMS regional system [17]. Three one-month visits in March 2025, October 2025 and May 2026, and the joint ^7Be implementation [7], establish the collaboration. Jesper Heile Christensen, Senior Researcher and a principal DEHM developer [24], will mentor model development, sensitivity design and interpretation, drawing on his data-assimilation experience [25].

Period / direction	Specific competence and verifiable output
M1–M4 AU → Chham	Source-label coding, boundary accounting and numerical verification with Ye: guided implementation, reviewed code and passing reconstruction/closure tests.
M4–M12 AU ↔ Chham	Structured training in inverse problems and uncertainty, coordinated by Ye: guided study of [8], synthetic-data exercises and a reviewed recovery/scoring protocol before fitting. Christensen mentors DEHM sensitivities. Output: reviewed protocol and a methods presentation.
M8–M18 Chham → AU	Radionuclide production, decay, collector conventions and sampling-aware R/Python workflows: two practical sessions with reusable exercises and unit tests.
M18–M24 Chham-led integration	Lead the held-out assessment, developer demonstration and open release; prepare a research portfolio and size-resolved ^7Be funding proposal with AU mentoring.

Table 5. Training is assessed through reviewed code, executable methods and independent scientific decisions.

Weekly meetings with Ye and quarterly reviews with Christensen support independent decisions. A career-development plan agreed in M1 is reviewed every six months. Within these meetings, Ye will guide reading, review synthetic-data exercises and assess uncertainty analysis; Christensen will review DEHM sensitivity design. Before fitting observations, Chham will present the recovery and scoring protocol for review. Chham leads implementation and analysis, and develops the follow-on proposal with AU mentoring.

Mentoring will use transparent authorship/contribution records, equitable opportunities to present and lead, gender-diverse participation in methods sessions, accessible hybrid materials and schedules compatible with caring responsibilities. Six-month reviews assess mentoring, workload and access to opportunities.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

Essaid Chham is a postdoctoral researcher at Universidad Miguel Hernández, with doctorates in Chemistry (Granada, 2018) and Physics (Abdelmalek Essaâdi, 2019), Erasmus Mundus experience and María Zambrano/APOSTD postdoctoral periods, detailed in Part B2. His combination of Monte Carlo modelling, environmental radioactivity and atmospheric analysis matches the project's integration of source physics, numerical tracers and observations.

His work on sampling intervals [11] informs the observation operator; Open-AMA [12] provides experience in reproducible tools for O2–O3. Research on ^7Be prediction [10] and DEHM, Fortran, R/Python and NetCDF skills support implementation. Teaching and supervision are detailed in Part B2. The fellowship adds controlled experiment design and uncertainty-based discrimination of competing explanations.

2. Impact

2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development

Within three to five years, Chham aims to lead an independent European research line on atmospheric transport and aerosols. By M24, he will lead the diagnostic release, target two lead-author manuscript submissions and prepare a follow-on proposal on size-resolved ^7Be in DEHM. AU mentoring supports its science, partnerships and budget for suitable calls, including NERD where eligible [26]. Six-month career reviews assess independent decisions, leadership and outputs. Results guide future observations requiring subsequent funding.

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities

Dissemination targets atmospheric modellers and environmental-radioactivity researchers; exploitation delivers a usable diagnostic method; communication explains why matching measurements can mask incorrect processes. Demonstrations, training and the data-access interface share the tested benchmark and release, within the existing WP5 allocation.

Audience / timing	Product and uptake measure
Scientific community M15–M24	Two manuscripts on multidecadal source/removal budgets and observational discrimination/transfer. The two manuscripts will target journals such as <i>Atmospheric Chemistry and Physics</i> and <i>Geoscientific Model Development</i> , according to their emphasis on atmospheric processes or reproducible model-evaluation methods. Preprints, open-access routes and at least one conference contribution. Track submissions, DOI releases and feedback.
AU/DEHM developers M12–M24	Demonstrate the benchmark at M12/M22 and deliver a developer note with supported changes and regression tests. Record feedback and reproduction effort; target at least one re-run by a DEHM/CAMS developer outside the project team by M24.
Monitoring/data specialists M18–M24	An observing-value brief on where paired deposition adds information and finer sampling helps. Invite feedback from JRC, DWD and relevant CTBTO/IMS specialists; record responses and reuse requests.
Wider research users / public M8–M24	An open example and data-access interface, two practical training sessions and an accessible radiotracer brief. Record attendance, tutorial completion and use of published outputs.

Table 6. Products, target audiences and measures of use guide dissemination and exploitation throughout the fellowship.

Reusable analysis tools will use an appropriate open-source software licence agreed with AU, respecting ownership and dependency compatibility; host-owned DEHM code remains separately governed. Versioned schemas, tests and a synthetic benchmark support independent reuse. A lightweight interface using established data-serving tools will provide direct downloads of published model subsets selected by period, area, variable and available hourly, daily or monthly resolution, with metadata, version and citation information. Curated products and publication-supporting data remain DOI-archived. Developer demonstrations and an independent re-run will assess usability and guide the final release.

2.3 The magnitude and importance of the project’s contribution to the expected scientific, societal and economic impacts

Scientific impact. STRAT-TRACE will link verified ⁷Be source budgets to observational discrimination of atmospheric explanations. Three wet settings with variable/fixed source forcing over 1986–2024/2025 test robustness across three complete solar cycles. Long European records add temporal reach; CTBTO air observations extend spatial tests within DEHM coverage. Paired concentration–deposition records determine where deposition adds information. The resulting map will identify compatible explanations and the additional sampling needed to distinguish them.

Technological and environmental-assessment pathway. DEHM is one of the 11 models contributing to the CAMS regional air-quality system [17]. Its developers will receive an executable benchmark, a reviewed demonstration and regression tests. Joint concentration–deposition tests will assess whether apparent gains in concentration skill remain consistent with deposition observations, and whether compensating supply–removal errors provide a plausible explanation. The results will guide model development. Usability, re-run effort and independent reproduction will be assessed through developer feedback. Transfer to pollutant forecasting remains a subsequent validation step.

Societal and European relevance. By helping DEHM developers test transport and removal biases, the benchmark can strengthen modelling evidence for EU air-quality assessments and the Zero Pollution Action Plan [18], subject to subsequent validation for relevant pollutants. The observing-value brief identifies where paired measurements or finer sampling add information. FAIR releases and reusable examples [19] support wider access. Environmental and economic benefits depend on adoption and validated pollutant applications.

3. Quality and Efficiency of the Implementation

3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

The 24-month plan links verified labels, matched experiments and independent observational tests. AU has confirmed resources and scheduling. MS1 at M4 checks budgets, source controls and input continuity before long runs. Reference output follows by M9 and the response library by M15.

WP / span / effort	Tasks and measurable output	Milestone
WP1 M1–M4 / 3 PM	Verify labels, boundaries and spin-up; audit inputs, observation coverage, overlaps and permissions; benchmark resources and lock evaluation blocks. D1: source/sampling protocol and data-management plan, including the historical-input contingency.	MS1, M4: numerical checks and source controls pass; common inputs, period and evaluation protocol fixed.
WP2 M3–M9 / 5 PM	Run R and its fixed-production partner over 1986–2024/2025 (78/80 configuration-years if separate). Assess solar and atmospheric conditions. D2: paired datasets and origin budgets.	MS2, M9: reference budgets verified; totals reconstructed.
WP3 M7–M15 / 6 PM	Run W–/W+ and fixed-production partners over the same years (156/160 configuration-years if separate); reweight sources and verify budgets. D3: response library and first manuscript draft.	MS3, M15: H1/H2 assessed; held-out cases retained.
WP4 M12–M21 / 5 PM	Run recovery and independent historical/recent tests; assess CTBTO geographic transfer with air records and joint skill on valid air–deposition pairs. Check stable segments and source controls. D4: discrimination map, observing brief and second manuscript draft.	MS4, M21: O3 classified, including explicit non-discrimination bounds.
WP5 M1–M24 / 5 PM	Training, supervision, data stewardship, developer engagement and dissemination. D5: licensed tools, data-access interface, methods materials, career portfolio and follow-on funding proposal.	MS5, M24: releases complete; two manuscript submissions targeted.

Table 7. Total effort: 24 person-months (PM) across overlapping work packages. WP2 starts with preparation; long runs follow MS1. The workload is for one researcher.

WP	1–3	4–6	7–9	10–12	13–15	16–18	19–21	22–24
WP1	●	●						
WP2	●	●	●					
WP3			●	●	●			
WP4				●	●	●	●	
WP5	●	●	●	●	●	●	●	●
Gate / output		MS1	MS2		MS3		MS4	MS5

Gantt chart. Filled cells indicate activity during all or part of each interval. D1–D5 correspond to MS1–MS5; the data-management plan is due in M3.

Execution budget and control of scope

Three wet settings over 39/40 years require 117/120 configuration-years with variable production. Fixed partners run as concurrent species where supported; separate runs double this to 234/240. AU's provision includes pre-1986 spin-up and a separate 20% restart/verification reserve. WP1 benchmarks wet/dry months in contrasting seasons with all 14 tracers, online budgets and input/output to refine scheduling and storage. All settings share one meteorological archive.

Delivery risks and scientific decision branches

The risk plan protects delivery while preserving the interpretation of inconclusive tests. A regime with weak discrimination still yields measured response limits and an observing-value assessment; inadequate information across the whole network would constrain the empirical contribution. The qualitative ratings below will be updated at each milestone.

Risk / residual likelihood / impact	Trigger, mitigation and retained result
Numerical closure or linearity fails Medium / High	Failed ledger or reconstruction tests stop source-fraction claims. Audit boundary, limiter and decay terms; use a verified coarser partition or source-restricted reruns within the costed reserve. Report the numerical limitation if closure cannot be established.
Paired records or metadata inadequate Medium / High	Audit JRC/DWD/CTBTO coverage, overlap, permissions and method eras. Retain valid pairs for joint tests and stable segments for temporal comparisons. CTBTO adds air tests; delivery delays do not displace the established paired baseline. Redesign if core paired coverage is inadequate.
Little discrimination across the available network Medium / Medium	Early synthetic tests evaluate the actual observing design. Report compatible scenarios and tested discrimination bounds; quantify the information added by alternative sampling. Preserve verified source-budget results and the geographic scope of inference.
Historical-input gaps or workflow delays Medium / High	If historical inputs cannot be completed by MS1, all wet settings and both source treatments use the longest verified common continuous period, prioritising the 2009–2024 observational baseline. Update solar-cycle coverage and evaluation blocks together. Checkpoints, online sums and the allocated reserve address workflow interruptions.
Candidate improvement fails held-out tests Medium / Medium	Separate air/deposition scores and source/discrepancy tests expose compensation. Deliver the rejected explanation and its regression test; retain the locked validation blocks.

Table 8. Qualitative residual assessments will be reviewed at each milestone. The response library and discrimination bounds remain deliverables when a particular explanation is rejected.

3.2 Quality and capacity of the host institutions and participating organisations, including hosting arrangements

AU's Department of Environmental Science develops atmospheric transport, chemistry and deposition models across scales [14,15]. Its DEHM contribution to CAMS [17] and the joint ⁷Be implementation [7] give the host the scientific expertise and developer connections needed for this project. The reduced-chemistry research configuration supports controlled diagnostics, with any later operational changes assessed in their relevant pollutant and forecasting configurations.

AU's confirmed provision supports the complete matrix and M9/M15 schedule. Chham will assemble the 1986–2024/2025 inputs, archive the CRAC-derived upper-boundary formulation and fields, and benchmark the allocated resources in WP1. Ye will oversee execution and scientific decisions. The signed CTBTO agreement complements JRC and DWD data access; WP1 verifies the usable records and permissions. Long runs follow MS1 source-control and input-continuity checks; historical contingencies retain a common period across configurations and source treatments.

Hourly receptor series and online pressure-band/regional budgets cover both source treatments over the full period; detailed 3-D fields are retained for selected episodes. At approximately 6 MB per hourly file, continuous output over 234/240 configuration-years would occupy 12–13 TB for that variable set. AU infrastructure provides over 600 TB of storage capacity. WP1 scales the estimate to retained tagged fields and episodes, budgeting meteorological inputs, checkpoints and copies separately. The M3 plan records TB requirements and retention, updated with pilot results. Curated daily/monthly products reduce download volumes, using time-weighted concentration means and integrated deposition; published subsets are directly accessible through the WP5 interface and repository.

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